RAML is a JSON-based schema embedded within research papers, either as inline annotations or in a separate metadata file. It enables authors and reviewers to track which sections of a manuscript were generated, modified, or influenced by AI.

RAML defines various levels of AI involvement: Generated, meaning the content was fully created by AI; Edited, where the content was generated by AI but reviewed and revised by a human; and Suggested, in which the content was suggested by AI but primarily authored by a human. RAML also includes metadata on the AI model used, such as GPT-4, Claude, or Gemini, along with version and parameters. Additionally, it supports version control by storing AI-generated drafts, change logs, and prompts used.

To encourage adoption, RAML should be incorporated into academic publishing standards such as ACM, IEEE, and Springer, which should establish guidelines requiring RAML annotations in AI-assisted papers. Preprint servers like arXiv and OpenReview should support RAML metadata for AI disclosure. Journals and conferences should require RAML compliance for submissions involving AI. Digital Object Identifiers (DOI) should include RAML information for research reproducibility.

RAML provides a structured approach to tracking AI contributions in research manuscripts, promoting transparency and accountability. As AI plays an increasing role in scientific discovery, standardized attribution mechanisms will be essential to maintaining the credibility and integrity of academic publishing.

3.8 Research Process Markup Language (RPML)

The AI Scientist faces difficulties in tracking detailed experimental metadata. Modifications made by its LLM-based coding assistant (Aider) are often undocumented, leading to reproducibility issues and making it hard to trace experimental changes.

We propose developing a standardized Research Process Markup Language (RPML) that captures all aspects of the research workflow in a structured format. RPML—based on a schema such as JSON or XML—should record experiment setups, code versions, container images, dataset versions, hyperparameters, and citation contexts automatically, ensuring full traceability and reproducibility.

3.9 Standardized Dataset Annotation and Integration

Tools like the AI Scientist rely on user-supplied datasets that often lack detailed, standardized metadata, limiting its ability to autonomously select high-quality relevant datasets for experiments.

We propose to encourage dataset providers (e.g., Hugging Face, Kaggle, UCI Machine Learning Repository) to adopt a standardized annotation schema. This schema should include comprehensive metadata such as domain applicability, data quality scores, versioning, benchmark results, and recommended use cases. With uniform metadata, the AI Scientist can automatically query and integrate datasets that best fit the experimental requirements.

3.10 Code Repositories Integration

The AI Scientist lacks access to recent baseline implementations from the state-of-the-art literature, making it challenging to benchmark its innovations against current best practices.

We propose to establish direct integrations with platforms such as PapersWithCode, GitHub, and Code Ocean. Standardized APIs and metadata from these platforms can be used to retrieve relevant source code repositories and performance metrics, allowing the AI Scientist to automatically download, run, and compare these baselines against its own results.

3.11 Participatory Strategic Retreat

The ideas presented in our paper should be seen as an initial stimulus for discussion. To explore the future of generative AI in IR research, key stakeholders in the community must be involved—particularly leading experts in the IR field, as well as young researchers whose future will be significantly shaped by generative AI. We believe that a strategic, multi-day workshop or retreat — such as ones held by the Dagstuhl or NII Shonan seminar series — would be the right setting for engendering an initial, larger-scale discussion where such in-depth discussions can inform the writing of white papers that codify opinions, similar to our opinions here. Such papers can then help catalyse the community to comment on and organize significant IR efforts, in practice and research.

4 Discussion and Conclusion

Our study confirms that the AI Scientist represents a significant step toward automating scientific research, yet our evaluation reveals that its current capabilities remain far from fulfilling its ambitious promises. While the AI Scientist can generate research ideas, it only sometimes successfully conducts experiments, and many of its generated ideas are not truly novel. It heavily depends on user input, struggles with methodological soundness, and lacks the ability to critically assess its own results. In its present form, the AI Scientist is, at best, an advanced research assistant that requires a lot of supervision, rather than an independent scientific agent.

A key takeaway from our study is that we believe AI systems like the AI Scientist will meaningfully contribute to the scientific critique process in the near future, benefiting both authors and reviewers. Researchers could leverage AI-generated feedback to iteratively refine their work through adversarial loops in which an AI challenges their hypotheses, methodologies, and analyses. Similarly, peer reviewers could use AI-assisted prompts to enhance their critiques, ensuring consistency and depth. Some academic conferences, such as ICLR, have already incorporated AI-generated review suggestions into their workflows. However, our findings indicate that AI-generated reviews by the AI Scientist often focus on surface-level critiques, while failing to detect deeper methodological flaws. This may make the job of second-level reviewers (such as area chairs or meta-reviewers) more important, as AI-generated reviews may lack depth but be well-formatted and directly address the review form's criteria. Such consolidators will have to look beyond surface-level analyses to judge whether a work holds promise.

Another promising role for the AI Scientist lies in replication and validation. Reproducibility is a cornerstone of scientific credibility, as highlighted by recent initiatives in recommendation systems (e.g., the Best Paper Award at RecSys) and NLP (e.g., ReproHum initiatives). AI could be employed to establish proof of work, ensuring that findings are replicable and that experimental procedures are verifiable. To achieve this, AI-driven replication efforts must prioritize transparency through open-weight and open-data models. Additionally, AI can support automated metadata creation and tagging (e.g., Dublin Core, Open Data Initiative) and facilitate structured data deposits in version-controlled repositories, providing verifiable provenance for scientific claims. These mechanisms incentivize transparent and reproducible research, potentially leading to new prestige metrics or blockchain-based certification systems for replication integrity.

Despite these potential benefits, integrating AI into scientific research presents ethical and practical challenges. AI models inherit biases from historical data and cannot independently distinguish between scientific quality and consensus. This limitation raises concerns that AI tools might reinforce outdated methodologies or amplify biases in peer review and publishing. Additionally, junior researchers may develop an overreliance on AI-generated suggestions, leading to automation bias that stifles innovation and independent thought. Research further indicates that AI can enhance scientific productivity, as shown in [17], yet its impact on researcher satisfaction remains complex. In one

study, while AI-assisted workflows increased efficiency, 82% of scientists reported lower job satisfaction afterwards. This paradox exemplifies a broader dilemma—if AI can outperform humans in key research tasks, what remains for human scientists to do? Addressing these risks requires AI-assisted workflows that actively encourage users to critically reassess assumptions and explore alternative perspectives.

One pressing concern is the potential for mass AI-generated paper submissions. Tools like the AI Scientist could enable researchers to generate large volumes of seemingly novel but low-value papers by tweaking existing codebases and experimental parameters. Such submissions could overwhelm academic venues, burdening already-overworked reviewers. The precedent set by AI-generated nonsense papers being accepted into conferences, such as those produced by SciGen, demonstrates the urgency of this issue. Unlike SciGen, modern AI-generated papers are often indistinguishable from human writing, making them harder to detect. Without proactive measures, the academic publishing ecosystem risks being inundated with research that adds quantity but not quality. To mitigate this risk, improved authorship verification, AI disclosure policies, and systematic detection methods must be developed.

The same risks extend to students using the AI Scientist to generate academic assignments. AI-generated essays, reports, and research papers could bypass the learning process, allowing students to submit work that appears well-structured but lacks genuine understanding or critical engagement with the subject matter. This raises significant concerns about academic integrity, as automated content generation makes plagiarism detection more difficult and blurs the line between acceptable assistance and unethical practices. Moreover, reliance on AI for coursework could hinder students' ability to develop essential skills in reasoning, analysis, and problem-solving. To address this, educators must adapt assessment strategies, incorporating oral defenses, iterative feedback loops, and AI-detection tools to ensure that students engage meaningfully with their work rather than merely outsourcing it to AI.

Looking ahead, the AI Scientist and similar tools represent an extraordinary technological leap. AI systems that once struggled to generate coherent text can now propose research ideas, design experiments, and draft manuscripts. While the AI Scientist currently operates at the level of an unmotivated undergraduate student, continued advancements could soon enable AI to produce research indistinguishable from that of dedicated Master's or PhD students. Future iterations may integrate real-time literature retrieval, deeper theoretical reasoning, and more rigorous experimental validation, progressively narrowing the gap between AI-driven and human-led scientific discovery. Recent developments such as Google's[22] and OpenAI's[23] *Deep Research* indicate that AI-driven research automation is rapidly advancing in this direction.

Finally, the Information Retrieval (IR) community must actively engage with these developments. The AI Scientist relies on IR techniques for literature search, data retrieval, citation analysis, and experimental design. As AI-driven research automation grows, the IR community has the opportunity to shape its methodologies. Addressing limitations in retrieval quality, document ranking, and automated citation analysis will be critical to ensuring AI-generated research is both credible and impactful.

In summary, while the AI Scientist falls short of its grand claims, it offers a glimpse into the future of AI-driven scientific discovery and towards 'Artificial Research Intelligence' (ARI). With responsible development and integration, AI could enhance knowledge generation, improve reproducibility, and streamline the research process in ways we are only beginning to explore. However, realizing this potential requires careful consideration of AI's limitations, ethical challenges, and long-term implications for the scientific community. The time to act and to participate in evaluating, exploring, discussing and developing AI-driven research agents is now!

[22] https://blog.google/products/gemini/google-gemini-deep-research/
[23] https://openai.com/index/introducing-deep-research/